

算术运算符

• ■ ○ / % **(乘方) //(商)

```
In [9]: a = 75
b = 20
print(a+b)
print(a/b)
print(a%b)
print(a//b)
print(a*b)
print(a-b)
print(b**2)
```

95
3.75
15
3
1500
55
400

比较运算符 bool True or False

```
In [62]: print(3==5)
print(3!=5)
```

False
True

赋值运算符 算术+等号 = += -= *= /= //= %= **=

```
In [7]: a = 10
# a += 5
# a *= 5
# a /= 5
print(a)
```

10

身份运算符: 比较两个变量的地址是否相同 is is not

```
In [6]: name1 = 'lihua'
name2 = 'xiaoyu'
print(id(name1))
print(id(name2))
print(name1 is name2)
print(name1 is not name2)
```

1519040522800
1519040514032
False
True

逻辑运算符 and or not 与或非

```
In [5]: x = True
y = False
```

```

z = False
print(x and y)
print(x or y)
print(y or z)
print(not y)

print(False and True)
print(0 and True)
print('a' and 'b')
print(1 or 0)

```

```

False
True
False
True
False
0
b
1

```

类型转换

```

In [10]: # 测试 ord(字符) → 数字
print("ord('a') =", ord('a'))
print("ord('A') =", ord('A'))

# 测试 chr(数字) → 字符 (只用32-126的可见字符)
print("chr(97) =", chr(97))
print("chr(35) =", chr(35)) # #号
print("chr(49) =", chr(49)) # 1

```

```

ord('a') = 97
ord('A') = 65
chr(97) = a
chr(35) = #
chr(49) = 1

```

```

In [4]: num = int(input("每天销售多少碗: "))
price = float(input("一碗面的价格: "))
days = int(input("每年营业多久: "))
total = num * price * days # 合计
print(f'销售额是:{total}')

```

销售额是:198000.0

变量的格式化输出 %s, %d, %f

```

In [31]: name = '66'
age = 18
cash = 1050.5
info = "名字是:%s, 年龄%d岁, 每月零钱2%f元"%(name, age, cash)
print(info)

```

名字是:66, 年龄18岁, 每月零钱21050.500000元

```

In [15]: name = input("输入姓名:")
phone = input("输入电话:")
company = input("输入公司:")
employ = input("输入职位:")
address = input("输入地址:")
print(f'姓名:{name} 电话:{phone} \n公司:{company} 职位:{employ} \n地址:{add

```

姓名:lihua 电话:6668888
公司:zhenshu 职位:jiaoshi
地址:wuhou

if条件句

```
In [24]: age = 20
         if age >= 18 :
             print("可以去网吧")
         else:
             print("不能去网吧")
```

可以去网吧

```
In [ ]: num = int(input("请输入一个数字: "))
         # 判断奇偶
         if num % 2 == 0:
             print(f"{num} 是偶数")
         else:
             print(f"{num} 是奇数")
```

9 是奇数

```
In [36]: a = float(input("请输入第一个数字: "))
         b = float(input("请输入第二个数字: "))
         c = float(input("请输入第三个数字: "))

         print(f"最大值是: {max(a, b, c)}")
         print(f"最大值是: {min(a, b, c)}")
```

最大值是: 10.0

最大值是: 5.0

```
In [38]: score = float(input("输入成绩:"))
         info = "你的成绩是:"
         if score >= 90:
             info += 'A'
         elif score >= 80:
             info += 'B'
         elif score >= 70:
             info += 'C'
         elif score >= 60:
             info += 'D'
         elif score <= 60:
             info += 'E'
         print(info)
```

你的成绩是:B

while循环语句

```
In [45]: i = 0
         while i <= 100:
             print(i)
             i += 1
```

0
1
2
3
4
5
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98
99
100

```
In [ ]: # 大写字母 A-Z
print("大写字母 A-Z:")
i = ord('A')
while i <= ord('Z'):
    print(f"{chr(i)} → {i}")
    i += 1

print("\n-----\n")

# 小写字母 a-z
print("小写字母 a-z:")
i = ord('a')
while i <= ord('z'):
    print(f"{chr(i)} → {i}")
    i += 1
```

大写字母 A-Z:

A → 65
B → 66
C → 67
D → 68
E → 69
F → 70
G → 71
H → 72
I → 73
J → 74
K → 75
L → 76
M → 77
N → 78
O → 79
P → 80
Q → 81
R → 82
S → 83
T → 84
U → 85
V → 86
W → 87
X → 88
Y → 89
Z → 90

小写字母 a-z:

a → 97
b → 98
c → 99
d → 100
e → 101
f → 102
g → 103
h → 104
i → 105
j → 106
k → 107
l → 108
m → 109
n → 110
o → 111
p → 112
q → 113
r → 114
s → 115
t → 116
u → 117
v → 118
w → 119
x → 120
y → 121
z → 122

```
In [60]: i = 1  
         sum_i=0
```

```
while i < 101:
    sum_i += i
    i += 1
print(sum_i)
```

5050

```
In [ ]: i = 2
sum_even = 0

while i <= 100:
    sum_even += i # 把当前偶数加到总和里
    i += 2        # 每次加2，直接跳到下一个偶数

print("1~100的偶数和是:", sum_even)
```

1~100的偶数和是: 2550

```
In [63]: line = ""
for _ in range(10):
    line += "*"
print(line)
```

作业1 求解1-100中能被3整除的所有数字的和

```
In [64]: s = 0
for i in range(1,101):
    if i % 3 == 0:
        s += i
print(s)
```

1683

作业2求解1-100 能被3和5同时整除的数字的和

```
In [66]: sum_i = 0
for i in range(1,101):
    if i % 3 == 0 and i % 5 == 0:
        sum_i += i
print(sum_i)
```

315

作业3求 $1 + 1/2 + 1/3 + 1/4 + 1/5$ 的和

```
In [67]: s = 0
for i in range(1,6):
    s += 1/i
print(s)
```

2.2833333333333333

作业4求 $1 + 1/2 + 2/3 + 3/4 + 4/5$ 的和

```
In [70]: s = 1
for i in range(2,6):
    s += (i-1)/i
print(s)
```

3.716666666666667

作业5输入一个数字将其转换为二进制，判断二进制中1的个数

```
In [74]: num = int(input("请输入一个数字: "))
count = 0
n = num
while n > 0:
    # 取n除以2的余数，如果是1，就计数+1
    if n % 2 == 1:
        count += 1
    # 把n整除2，相当于右移一位
    n = n // 2

print(f"数字 {num} 的二进制中,1的个数是:{count}")
```

数字 15 的二进制中,1的个数是:4